



US Commercial Aviation Operational Performance Report

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Dataset: 2015 Bureau of Transportation Statistics (BTS) On-Time Reporting (5.82M Rows)

1. Introduction

The purpose of this report is to diagnose systemic operational bottlenecks, seasonal inefficiencies, and airport-level performance trends within the United States commercial aviation network. Utilizing a massive historical dataset containing approximately 5.82 million domestic flights, this analysis establishes baseline operational KPIs, isolates root causes of flight delays, and evaluates carrier performance. The ultimate goal is to provide data-driven strategic recommendations to optimize network reliability and improve the passenger experience.

2. Methodology

To process, analyze, and visualize this high-volume dataset, a robust, end-to-end data pipeline was engineered across the SQL database, Power Query transformation layer, and the front-end Power BI data model:

Phase 1: Database Engineering & SQL Optimization

Before ingesting data into the BI layer, analytical optimizations were performed directly inside the relational database management system to maximize performance and handle the massive 5.82M row layout:

- **View Creation (v_analytical_flights):** Instead of pulling raw, heavy database tables into Power BI, a customized database view was created. This view joined the primary flights table with carrier lookups, handling structural transformations at the database level to ensure data immutability and high query performance.
- **Pre-Aggregation & Validation:** High-level SQL queries were written to calculate global baselines (such as total flight volume and average arrival/departure times) to establish a trusted data benchmark for validating downstream BI metrics.

EDA & KPI's Definition(SQL):

- 1) Overall flight volumes, cancellations (total, by reason), and diversions

	total_flights	total_cancellation	total_diversions
▶	5819079	89884	15187

2) Basic statistics for departure and arrival delays (average, median, min, max)

	avg_departure_delay	min_departure_delay	max_departure_delay	avg_arrival_delay	min_arrival_delay	max_arrival_delay
▶	9.23	-82	1988	4.33	-87	1971

3) The distribution of different types of delays (airline, weather, NAS, etc.)

	airline	weather	nas	late_plane	security
▶	20172956	3100233	14335762	24961931	80985

CORE KPI CALCULATION:

1) On-Time Performance (OTP) Rate (%):

	otp_rate_pct
▶	80.87

2) Average Delays in arrival and departure:

	KPI_avg_departure_delay_mins	KPI_avg_arrival_delay_mins
▶	9.23	4.33

3) Cancellation Rate (%):

	cancellation_rate_pct
▶	1.54

Phase 2: Data Extraction & Transformation (Power Query)

Once the optimized SQL view was established, it was ingested into Power Query for final dimensions mapping and cleaning:

- **Advanced M-Code Mapping:** Crucial text-translation logic was implemented to map numerical identifiers into human-readable business dimensions. This included translating raw airport codes (e.g., mapping 14747 to "SEA", 14771 to "SFO", and 14222 to "IFP") and calendar values (1 through 7 to "Monday" through "Sunday"). This step removed analytical confusion and standardized the visuals.

Phase 3: Relational Data Modeling & DAX Engineering

Data was loaded into a high-performance semantic data model inside Power BI to prepare for interactive calculation states:

- **Calculated Columns:** A new dynamic calendar attribute (Flight Hour) was constructed using structural date-time extractions to segment flight distributions cleanly into continuous 24-hour cycles.
- **Core KPI Measures:** Key performance metrics were explicitly calculated using Data Analysis Expressions (DAX) to guarantee real-time precision during dynamic filtering. Most notably, the gold-standard operational metric, **On-Time Performance (OTP) Rate**, was engineered to safely handle blank calculations and isolate non-cancelled flights delayed by 15 minutes or less:

```
OTP Rate % =  
DIVIDE(  
    CALCULATE(  
        COUNT('airline_performance v_analytical_flights'[FLIGHT_NUMBER]),  
        'airline_performance v_analytical_flights'[DEPARTURE_DELAY] <= 15,  
        'airline_performance v_analytical_flights'[CANCELLED] = 0  
    ),  
    COUNT('airline_performance v_analytical_flights'[FLIGHT_NUMBER]),  
    0  
)
```

3. BI Dashboard Architecture Overview

The interactive portal is structured into dedicated analytical modules designed for executive decision-making:

1. **Landing Page (Portal Anchoring):** A high-impact executive interface framing the scope of the project (5.8M flights analyzed across 14 commercial carriers) with direct user-navigation pathways.
2. **Executive KPI Summary:** A macro-review page leveraging heavy-metric callout cards for core KPIs (Total Flights, OTP, Cancellations), a dynamic monthly timeline slider, and a categorical "Day of Week" tile slicer. This page features a Donut Chart breaking down cancellation reasons and a horizontal bar chart benchmarking carriers.
3. **Trend Analysis (Temporal Trends):** A page dedicated to deep-diving systemic volatility. It maps average arrival delays across continuous months and utilizes a 24-hour continuous line graph to highlight the late-afternoon operational degradation.
4. **Airport Performance:** A geospatial and structural layout mapping the top performing and worst performing nodes. It combines a geographical bubble map of the United States with localized performance bars to help regional managers isolate infrastructure vulnerabilities.

4. Core Findings

Based on the data synthesis, several critical operational insights were uncovered:

- **Network Baseline:** The front-end interactive visualizations establish an **80.99% On-Time Performance (OTP) Rate**, validating tightly against the raw, backend SQL view base baseline of **80.87%**. The overall flight cancellation rate is tightly managed at **1.54%**. When delays occur, the macro-network averages a **9.23-minute departure delay** and a **4.3-minute arrival delay**.
- **Temporal Inefficiencies:** Operational volatility heavily correlates with specific calendar months and times of day. **June** represents the worst operational month for delays, while **October** emerges as the most stable and efficient.
- **The Hourly Rush-Hour Effect:** Daily delay analysis revealed a highly predictable continuous trend line. Flight delays hit their absolute lowest point in the early morning (05:00 hours) and steadily compound throughout the day, peaking dramatically at **19:00 hours (7:00 PM)** with an average arrival delay spike of **9.9 minutes**.
- **Severe Airport Bottlenecks:** Laughlin/Bullhead International Airport (**IFP**) represents the most severe bottleneck in the structural network, exhibiting the highest average departure delay (**89.11 minutes**) and the worst local OTP. Conversely, Ithaca Tompkins Regional (**ITH**) exhibits the highest cancellation rate in the country at **11.76%**, primarily driven by National Air System (NAS) and weather disruptions.

5. Strategic Recommendations

To improve network reliability and lift the national OTP rate above the 80.99% baseline, the following actions are recommended:

- **Implement Late-Afternoon Throttle Strategies:** Because delays systematically snowball and peak at 19:00 hours due to consecutive aircraft routing delays, airlines should introduce tactical block-time buffers for flights scheduled between 16:00 and 20:00 hours to absorb cascading delays.
- **Targeted Regional Infrastructure Intervention:** Prioritize immediate operational reviews at regional nodes like **IFP** to diagnose why departure delays average a staggering 89 minutes. Modifying gate availability or runway scheduling at these specific micro-hubs will prevent localized disruptions from bleeding into larger major hubs.
- **Expand Seasonal Readiness Programs:** Given that June is established as the lowest-performing operational month for delays due to summer traffic, and weather accounts for over 54% of network cancellations, regional allocations for high-traffic summer scheduling buffers and Q1 winter de-icing infrastructure should be aggressively expanded.

